Justin Spenker

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PROFESSIONAL SUMMARY

Computer Engineering student at UC Davis with hands-on experience spanning PCB design and layout, real-time embedded software, FPGA-based digital design, and clean room semiconductor fabrication. Proven ability to take hardware systems from schematic through bring-up and physical validation, including SMD soldering, functional testing, and iterative debug across multiple academic project contexts.

EDUCATION

University of California, DavisDavis, CA
B.S. Computer Engineering (In Progress), Expected Dec 2026

PROJECTS

Autonomous Lane-Following Vehicle
Software Lead

- Designed and implemented real-time lane-following software in Python (MicroPython) on an OpenMV camera, splitting the frame into multiple zones with independent linear regression on each side and a center-weighted average to define a target corridor with configurable error tolerance
- Designed and tuned a PD (proportional-derivative) controller for closed-loop steering with dynamic speed control, adjusting motor PWM based on steering demand and curve detection
- Engineered the camera mounting system to maintain consistent field of view across track surfaces, directly supporting reliable zone detection and regression accuracy
- Collaborated on a custom two-layer motor control PCB in Altium Designer integrating a VNH5019 H-bridge motor driver, 5V/6V voltage regulators, servo/OpenMV connectors, and star-ground routing for noise isolation
- Hand-soldered SMD and through-hole components; diagnosed and resolved solder bridging on the VNH motor driver using solder wick, flux, and manual rework; validated all connections via continuity testing

FPGA Fault Simulation for Specialized Hardware

- Reverse-engineered a Tank Battalion arcade PCB schematic, tracing horizontal/vertical timing logic (HSYNC, HBLANK, VSYNC, VBLANK), 6502 CPU control signals, and analog sound generation paths
- Assumed full ownership of the project after both teammates dropped the course, independently completing all fault-injection design, testing, and documentation originally scoped for a 3-person team
- Designed and executed several dozen targeted fault-injection experiments covering stuck-at-0/1, single-event upset/bit flip, and simulated capacitor failure on a DE10-Nano FPGA using the MiSTer platform; exploited the framework's lack of boot security to directly modify HDL modules and recompile via Quartus Prime for each test iteration
- Targeted sprite rendering, sound amplifier, and physics/collision subsystems based on observability through native CRT video and audio output; produced gameplay-breaking symptoms including wall-clipping collision failures, missing weapon audio, and non-firing weapon logic in over half of tested cases
- Identified that HSYNC/VSYNC and certain HBLANK/VBLANK faults were silently auto-corrected by the MiSTer framework and carried risk of physical CRT damage; narrowed test scope accordingly to protect hardware while continuing investigation on safer subsystems
- Authored an IEEE-format technical report documenting fault-injection methodology, observed subsystem behavior, and root-cause findings

Quantum Key Distribution Protocol Simulation

- Simulated qubit transmission across variable distances and measured protocol fidelity and success rate using Qiskit
- Analyzed the effect of noise models on transmission reliability compared to ideal models

Semiconductor Fabrication Study

- Completed a university semiconductor fabrication research study, gaining practical exposure to clean room protocols including ISO gowning procedures, contamination control, and general wafer processing steps

EXPERIENCE

Shift LeaderMay 2024 – Present
TTL Automotive — Davis, CA

- Supervised and motivated a team of 23 employees to deliver fast, high-quality service in a high-volume environment
- Oversaw daily operations including chemical checks, task delegation, and opening/closing procedures
- Guided customers through wash selection and safety guidelines, ensuring smooth and efficient operations

Independent Sales RepresentativeSep 2022 – Jun 2024
2020 Companies (Samsung & HP at Best Buy) — Vacaville, CA

- Represented Samsung and HP within Best Buy locations, driving brand awareness and product sales through direct in-store engagement
- Trained and coached Best Buy associates to strengthen product knowledge and sales confidence on key technology items
- Delivered personalized technology recommendations to customers, consistently meeting sales targets

SKILLS

Hardware Design: PCB layout & routing (Altium Designer), FPGA design (Quartus Prime), schematic capture, DRC/DFM validation

Embedded Systems: OpenMV (MicroPython), PWM/servo control, real-time control loops, motor driver interfacing, sensor integration

Fabrication & Lab: SMD & through-hole soldering, oscilloscope, multimeter, logic analysis, clean room protocols (ISO gowning), semiconductor fabrication fundamentals

Fault Analysis & Verification: fault injection (stuck-at, SEU, component failure), HDL-level debug, subsystem isolation, ModelSim, IEEE technical reporting

Programming: Python (MicroPython, Qiskit), HDL (RTL & Verilog), PowerShell

Other: Network fundamentals (DNS, VPN, TCP/IP), Windows systems, technical sales & customer communication